

Basic DSP

Lowpass filter design `[B1, A1] = butter(order, fc/fnyq);`

Highpass filter design `[Bh, Ah] = butter(order, fc/fnyq), 'high');`

Filtering a signal *u* `uf = filter(B1, A1, u);`

Autocorrelation (discrete)
$$\varphi_{xx}(\tau) = \sum_{k=0}^{N_{acf}-1} x(k) x(k + \tau) , \quad 0 \leq \tau \leq N_s - N_{acf}$$